

RAG & Runbook Knowledge Architecture

ServiceGuard AI

1. Knowledge sources

- Production incident runbooks.
- Database failure procedures.
- API degradation procedures.
- Infrastructure escalation policy.
- SLA policy.
- Customer communication policy.
- Disaster recovery procedures.

2. Ingestion

```
Document
↓
Parser
↓
Structure-aware chunking
↓
Metadata
↓
Embedding
↓
pgvector
↓
Retrieval index
```

3. Metadata

- document_id, title, version, effective_date, service_type, incident_type, section, source, permissions.

4. Retrieval policy

- Filter by effective/current version.
- Use semantic similarity plus metadata filters.
- Retrieve a bounded number of chunks.
- Return evidence IDs and source metadata.
- Reject stale or unauthorized documents.

5. Grounding rules

- The LLM must distinguish retrieved policy from live telemetry.
- No evidence means no invented procedure.
- Recommendations should reference the runbook evidence used.
- Conflicting runbooks require explicit escalation to the latest approved source.

6. Initial implementation

Use PostgreSQL + pgvector first. Pinecone remains an optional scale-out choice rather than a mandatory dependency for MVP.